

SHM Worked Example 5.15

Reference: Applied Mechanics: Reed's Vol. 2

Ex. 5.15: The stroke of a reciprocating engine is 500 mm. Neglecting the effect of the angularity of the connecting rod, find the velocity and acceleration of the piston when the crank is 30° past top dead centre and the engine is running at 750 rpm.

Calculations

1. **Determine the Radius of the Crank:**

$$r = \frac{500}{2} = 250 \text{ mm} = 0.25 \text{ m}$$

2. **Angular Velocity of the Crank:**

$$\omega = \frac{2\pi \cdot 750}{60} = 78.54 \text{ rad/s}$$

3. **Velocity of the Piston:**

$$V_p = \omega \cdot r \cdot \sin \theta = 78.54 \times 0.25 \times \sin(30^\circ) = 9.82 \text{ m/s}$$

4. **Acceleration of the Piston:**

$$a_p = \omega^2 \cdot r \cdot \cos(\theta) = 78.54^2 \times 0.25 \times \cos(30^\circ) = 1335.52 \text{ m/s}^2$$